Memories in Scaled technologies: A Review of Process Induced Failures, Test methodologies, and Fault Tolerance*

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***Abstract*— The inter-die and intra-die variations in process parameters (in particular, threshold voltage (Vt)) can lead to large number of failures in an SRAM array, thereby, degrading the design yield in nanometer technologies. To improve parametric yield of nano-scaled memories, different circuit and architectural level techniques can be used. In this paper, we first analyze and model different SRAM failures due to parameter variations, and discuss test methodologies to test for process variation induced failures. Next, we describe two different self-repairing techniques–at the circuit level, using adaptive body biasing and at the architecture level, using Built-in-self-test (BIST), redundancy and address remapping. The discussed self-repair mechanisms can improve design yield much beyond what can be achieved using row/column redundancy and error correcting codes (ECC) alone.**

I. INTRODUCTION

Die-to-die and within-die variations in process parameters result in the mismatch in the strengths of different transistors in an SRAM cell (Fig. 1), which results in functional failures (read, write, access and hold failures). The functional failures due to parametric variations (hereafter, referred to as parametric failures) degrades the memory yield (i.e. the number of non-faulty chips) [2]. The principal reason for parametric failures is the intra-die variation in threshold voltage of the cell transistors due to random dopant fluctuations [1-2]. However, the die-to-die variation in process parameters (say, *Vt*) also has a strong impact on the failure probability of a cell. In particular, low-Vt dies suffer mostly from the read and hold failures while high-Vt dies suffer from access and write failures [2]. Thus die-to-die variations significantly degrade the yield. Hence, there is a need to model parametric failures induced by process parameter variations and develop test methodologies to improve fault coverage.

In addition, conventionally, redundant rows/columns [5-6] are used in memories to improve yield. These redundancy techniques have limitation on the number of faulty rows/columns it can handle, due to resource limitation and design complexity. In particular, the failures due to random dopant effect are randomly distributed across the dies, resulting in a large number of faulty rows/columns. Recovery from such defects is difficult to handle by row/column redundancy alone. In this paper, we will show that, with the drastic increase in parameter variation, redundancy alone will not result in good yield. Moreover, SRAM failures due to process variation change depending on operating condition (e.g supply voltage, frequency). The operation condition changes dynamically, which poses challenges to any static techniques, such as row/column redundancy. ECC [7-8], employed in existing design to correct transient faults (soft error), can also be used to correct failures due to process variations. However, ECC has limitations on the number of error bits it can correct. Also, ECC is implemented with considerable area and performance overhead.

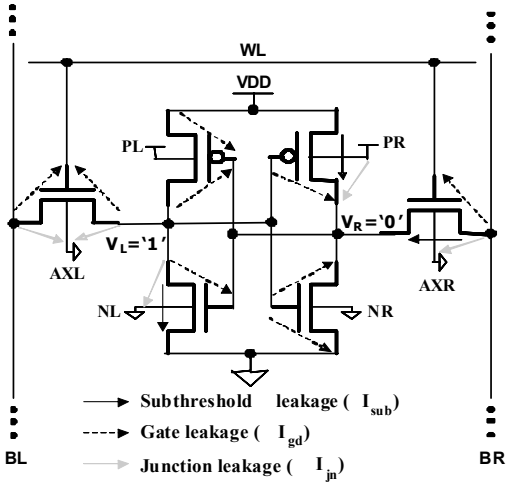


Fig. 1: SRAM Cell storing "0" at node R.

With the limitation of the existing fault tolerant scheme, SRAM, that can repair itself and, hence, reduce the number of failures would be very helpful for yield improvement in nano-scale technologies. In this paper, we consider two self-repairing techniques at the circuit and architecture level: a) A circuit technique that senses the process corner of the die and applies adaptive body bias to improve yield; b) An architecture technique that uses Built-In-Self-Test (on-line or off-line) to determine a failure map of the memory, re-maps the address based on the failure map, and improves yield at the cost of little performance overhead.

Adaptive Body Bias (ABB) techniques, which are traditionally used for logic circuits [3-6], can be used as an adaptive repair technique for memory. Application of Reverse-Body-Bias (RBB) in low-Vt dies increases their Vt thereby reducing possible read/hold failures in SRAM cells present in them [2]. Similarly, application of Forward Body Bias in high-Vt dies decreases their Vt, which reduces the access and write failures in the SRAM cells. However, major obstacle in the application of the adaptive repair techniques in memory is the presence of large intra-die variation. Due to high intra-die variation it becomes difficult to distinguish between a die from low Vt (inter-die) process corner and a die from high Vt (inter-die) process corner. Hence, separation of the

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dies in different inter-die Vt corners (hereafter, referred to as Vt-binning) is very important for the application of adaptive and self-repair techniques.

The proposed SRAM design techniques significantly reduce the number of failures in an SRAM array. The proposed design is implemented in BPTM 70nm technology and simulated in HSPICE. Our analysis shows that the self-repair technique in the SRAM improves the yield by 5%-40% depending on the inter-die and intra-die Vt variations. We have fabricated the self-repairing SRAM using adaptive body-biased SRAM in IBM 130nm technology to show the self-repair operation.

II. SRAM Parametric Failures and Test Optimization

A. Parametric Failures in SRAM Cell and Array

Increasing inter-die statistical variations in the process parameters (channel length (L), width (W), and transistor threshold voltage (Vt)) has emerged as a serious problem in the nano-scaled circuit design [1]. The inter-die parameter variations, coupled with the intrinsic on-die variation in Vt due to random dopant fluctuation, can result in failure of SRAM cells [1]. A cell failure can occur due to: (a) an increase in the cell access time (access failure), (b) unstable read/write operations (read/write failure), or (c) failure in the data holding capability of the cell at a lower supply voltage (hold failure in the stand-by mode). A failure in any of the cells in a column (or row) of the memory will make that column (or row) faulty. If the number of faulty columns (or rows) in a memory chip is larger than the number of redundant columns (or rows), then the chip is considered to be faulty. Hence, the failure probability of a cell is directly related to the yield of a memory chip. Consequently, estimation of the failure probability for a cell is necessary in the design phase to ensure a good yield. We have developed a method to predict the yield of a memory chip under inter-die variation of L, W and Vt by estimating the failure probability of the cell considering intra-die parameter variation. The method is developed considering intra-die Vt variation (principally due to random dopant fluctuation) and can be extended to include on-die random variation in L and W.

The intra-die Vt variation ((Vt) due to random dopant fluctuations (RDF) results in failures in SRAM cell. Fig. 1 shows the schematic of an SRAM cell storing "0" at node R. The (Vt's of the cell transistors due to RDF, can be considered as independent Gaussian random variables with standard deviation given by [8]:

$$\sigma_{\delta Vt_i} = \frac{qT_{ox}}{\varepsilon_{ox}}\sqrt{\frac{N_{SUB}W_{dm}}{3LW}} \quad (1)$$

where, Tox is the oxide thickness, W_{dm} is the width of the depletion region, and N_{SUB} is the doping concentration in substrate. The parametric failures in an SRAM cell are principally due to [8]:

Read Failure - Flipping of the SRAM cell data while reading. The read failure can be reduced by increasing the difference between the voltage rise at the node storing "0" while reading (say, V_{READ}) and the trip-point of the inverter (V_{TRIPRD}) associated with the node storing "1".

Write Failure – Unsuccessful write to the SRAM cell. Write failure occurs if the node storing "1" cannot be discharged through the access transistors during the word-line turn on time.

Access Failure –Access failure occurs if the voltage differential between the two-bitlines is below the offset voltage of the sense-amplifier at the time of the sense-amplifier firing [15]. Access failure occurs due to the reduction of the bit-line discharging current through the access and pull-down NMOS transistors.

Hold Failure - The destruction of the cell data in the standby mode with the application of a lower supply voltage. The hold failure occurs due to high-leakage of the NMOS transistors connected to the node storing "1". At a lower VDD, due to the leakage of the NMOS, the node storing "1" reduces from VDD (which is enhanced by a weak PMOS). If that voltage becomes lower than the trip-point of the inverter storing "0" the cell flips in the standby mode.

If a cell in a column fails, the column is replaced by an available redundant column (column failure). If the number of faulty columns is larger than the number of redundant columns, the SRAM array fails (memory failure). Given the failure probability of the SRAM cell (P_F), the column (P_{COL}) and the memory failure probabilities (P_{MEM}) are estimated as:

$$P_{COL} = 1-(1-P_F)^N;\ P_{MEM} = \sum_{i=N_{RC}+1}^{N_{COL}} \binom{N_{COL}}{i} P_{COL}^{i}\left(1-P_{COL}\right)^{N_{COL}-i} \quad (2)$$

B. Test Techniques to Detect Failures

March test is prevalently applied to SRAM test [18]. To cover the traditional single cell and coupling fault models, March C- is popularly used as a base sequence [18]. However, a serious problem with March C- is that it does not detect deceptive read destructive faults, which are very likely to happen in memories due to read failures. In [19], a test sequence named March SR is proposed that covers the deceptive read destructive faults. However, March SR ignores address decoder faults. Furthermore, the sequence has a test time of 14N (N being the number of addresses in a memory), which is significantly higher compared to the test time of March C- (10N). Moreover, conventional March test sequences do not cover faults resulted from hold failures. However, it is increasingly important to test for these faults because of the necessity to apply low supply voltage in the standby mode to reduce leakage power consumption in scaled technologies. Due to Vt mismatches resulting from intra-die variations, lowering supply voltage induces hold failures in SRAM cells. Hence, modification of the March sequences for failures induced by process variations is an emerging requirement. We proposed two March sequences that can cover the failures due to process variations with minimum impact on test time.

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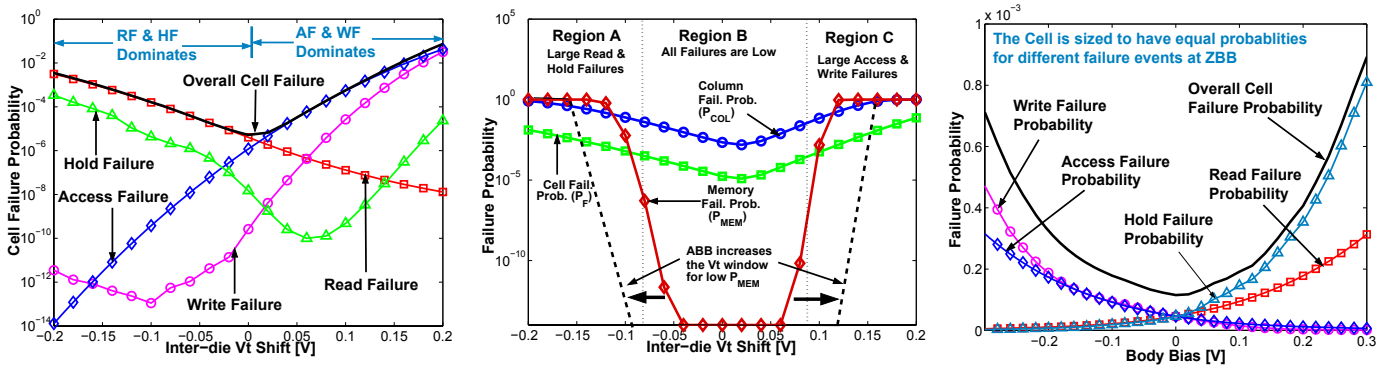


Fig. 2: Effect of inter-die Vt shift and body-bias on the failure probabilities: (a) cell failure probability with inter-die Vt shift, (b) memory failure probability with inter-die Vt shift, and (c) effect of body-bias on cell failure.

The first sequence is based on the well-known March C-. We have observed that two extra read operations are required to detect read failures. In addition, for detection of hold failures, proper places are identified in the sequence (denoted by HOLD) to lower the supply voltage; keep the memory at lower supply for a specified time (at least one cycle); and then bring it back to normal supply. The March C- sequence with the above-mentioned modifications is presented as follows:

Optimized March C- : c (W0) ⇑ (R0 W1) ⇑ (R1 W0) ⇓ (R0 W1) (HOLD) ⇓ (R1 R1 W0) (HOLD) c (R0 R0)

We refer to this proposed sequence as Optimized March C-. The optimized March C- has a test time of 12N, resulting in an improvement of 15% compared to March SR [4]. We assume that the time required for HOLD is insignificant compared to the time required for one March operation to all the addresses. Furthermore, the optimized March C- has better fault coverage than March SR [20].

In order to maintain the test time of March C- (10N) and yet detect both read failures and hold failures, we also developed a novel test sequence, namely March Q, as described below:

March Q : ⇓ (W0) (HOLD) ⇑ (R0 W0 W1 R1) (HOLD) ⇑ (R1 W1 W0 R0) ⇓ (R0)

It was observed that the optimized March C- sequence has the best fault coverage, with a test time increase of 20% compared to March C-. March Q detects the deceptive read destructive faults, while maintaining the shortest test time (10N). However, March Q can statistically detect only half of the transition coupling faults and is not capable of detecting address decoder faults (like March SR) [20]. It is shown that, with a DFT circuit, the test time of optimized March C- can be reduced to 10N and all the faults due to process variations are covered [20].

III. Self-Repairing Using ABB

With the above-mentioned model for SRAM failures and test techniques, memory failures can be estimated in an early design phase and memory test can be applied in a time efficient manner. Moreover, to further improve the memory yield, we propose two self-repairing memory design techniques in the following two sessions. In this session, we introduce a circuit technique which applies the adaptive body bias (ABB) to SRAM based on monitoring the SRAM leakage, which shows the process corner the memory chip sits in. We have shown that through correct body biasing, the yield of SRAM can be improved significantly.

A. *Effect of Inter-die Vt shift on Cell Failures*

A Vt shift toward low Vt process corners, due to inter-die variation, increases the read and the hold failures of SRAMs (Fig. 2a). This is because of the fact that, lowering the Vt of the cell transistors increases V_{READ} and V_{TRIPRD}, thereby increasing read failures [21]. The negative Vt shift increases the leakage through the transistor N_L, thereby, increasing the hold failures. On the other hand, for SRAM arrays in the high-Vt process corners, the probabilities of access failures and write failures are high (Fig. 2a) [21]. This is principally due to the reduction in the current drive of the access transistors. The hold failure also increases at the high Vt corners, as the trip-point of the inverter PR-NR increases with positive Vt shift. Hence, the overall cell failure increases both at low and high-Vt corners and is minimum for arrays in the nominal corner (Fig. 2a). Consequently, the probability of memory failure (Eq. 2) is high at both low-Vt and high-Vt process corners (Fig.2b).

B. *Effect of Body-bias on Cell Failures*

Let us now discuss the effect of the body-bias (applied only to NMOS) on different types of failures. Application of reverse body-bias increases the Vt of the transistors which reduces V_{READ} and increases V_{TRIPRD}, resulting in a reduction in the read failure (Fig. 2c) [21]. The Vt increase due to RBB also reduces the leakage through the NMOS thereby reducing hold failures (Fig. 2c). However, increase in the Vt of the access transistors due to RBB increases the access and the write failures. On the other hand, application of FBB reduces the Vt of the access transistor which reduces both access and write failures. However, it increases the read (V_{READ} increase and V_{TRIPRD} reduces) and hold (leakage through NMOS increases) failures (Fig. 2c) [21].

C. *Application of Adaptive Body Bias to Enhance Yield*

From Fig. 2c, it can be observed that, above a certain Vt-shift (~100mV) small changes in inter-die Vt results in a large memory failure probability (regions A and C). However, for chips with Vt in the window of -100mV to 100mV (region B) the memory failure probability (P_{MEM}) is negligible. Using the memory failure

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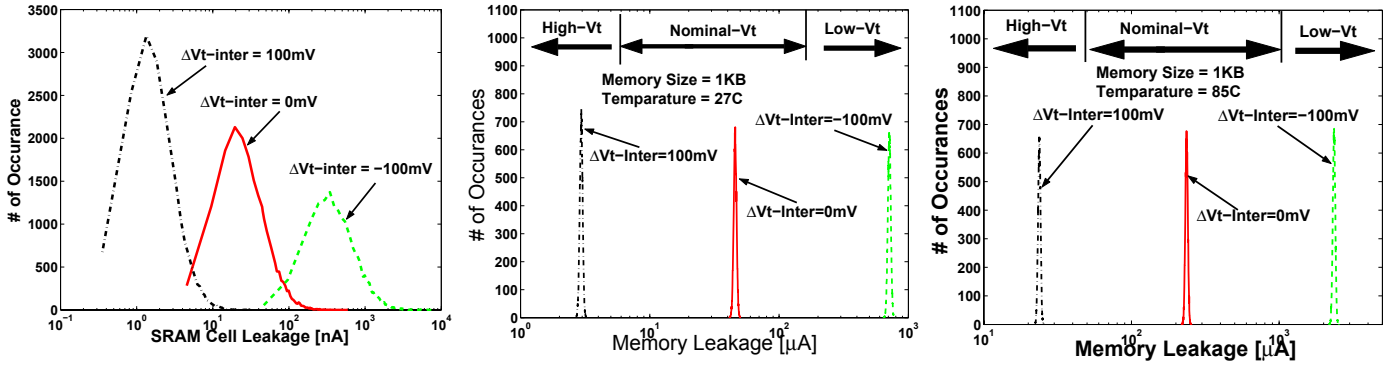


Fig. 3: Effect of random intra-die Vt variation at different inter-die Vt corners of SRAM: (a) leakage distribution (due to intra-die variation) of an SRAM cell, (b) leakage distribution (due to intra-die variation) of the 1KB SRAM array at T=27C, and (c) at T=85C.

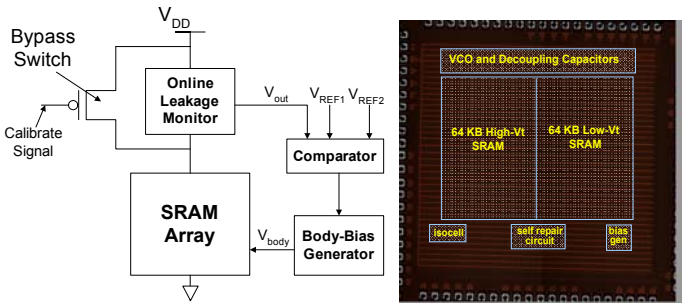


Fig. : Self-re airing SRAM scheme and die- hoto.

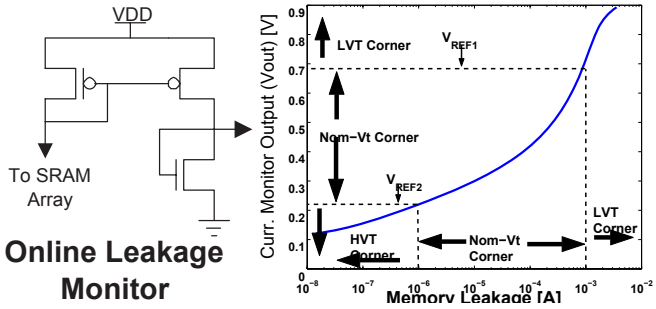


Fig. 5: Current mirror circuit and characteristics

probability the yield of the memory can be defined as [2, 8]:

$$Yield = 1 - \left(\sum_{INTER=1}^{N_{INTER}} P_{MEM}\left(Vt_{INTER}\right) \Big/ N_{INTER} \right) \quad (3)$$

where, N_{INTER} is the total number of dies. Let us further assume that the number of dies in region A, B, and C are N_A, N_B and N_C, respectively. Hence, yield can be obtained as:

$$Yield = 1 - \left(\frac{P_A N_A + P_B N_B + P_C N_C}{N_A + N_B + N_C} \right) = \frac{N_B}{N_A + N_B + N_C} \quad (4)$$

where, P_A(~1), P_B(~0), and P_C(~1) are the memory failure probabilities in the region A, B and C. Hence, to improve yield, N_A and N_C have to be reduced (in other words N_B needs to be increased). This can be achieved by applying RBB to the dies in region A thereby reducing their read and hold failure probability. Similarly, application of FBB to the chips in region C reduces their write and access failure probability. This effectively, increases the ΔVt(inter) window for region B resulting in a higher value for N_B and lower values for N_A and N_C. Hence, adaptive application of the body bias (Adaptive Body Bias, ABB) based on the inter-die process corner of a die, can improve the memory yield [21].

D. Vt-binning with Leakage Monitoring

To determine the correct body bias to apply to the SRAM chip for failure probability improvement, the process corner, in which the memory chip sits, needs to be determined. Here, a Vt-binning technique based on leakage monitoring is proposed. The leakage of an SRAM cell is composed of the subthreshold, the gate and the junction tunneling leakage as shown in Fig. 1 [2]. The random intra-die variation in threshold voltage results in significant variation in cell leakage, particularly, the subthreshold leakage. Considering Vt variation in different transistors, the total subthreshold leakage of the SRAM cell can be expressed as [7]:

$$I_{sub} = \sum_{i=AXR,NL,PR} W_i I_{sub0-i} exp\left(\frac{-\delta Vt_i}{m_i\, kT/q} \right) \quad (5)$$

where, $I_{sub0\text{-}i}$ represent the subthreshold leakage of the i-th transistor (with width of 1μm) at V_{gs}= V_{bs} =0, V_{ds}=V_{DD}. W_i, δVt_i, and m_i represent the width, Vt shift from zero and subthreshold swing factor of the i-th transistor. Considering the exponential dependence of the subthreshold leakage on Vt and using Wilkinson's method, the total subthreshold leakage of the cell be modeled as a Log-Normal variable with the mean and the standard deviation obtained using [9, 10]. The gate and the junction leakage only increase the mean as they does not strongly depend on Vt. The leakage of an SRAM array can be obtained by adding the leakage of all the cells (say, N_{CELL}) in the memory. The major source of intra-die Vt variation in SRAM array is the RDF. Since RDF induced Vt variation is completely random, the leakage of different cells can be considered as independent random variables. Hence, using the overall memory leakage (L_{MEM}) can be assumed to be Normal with mean (μ_{MEM}) and the standard deviation (σ_{MEM}) given by:

$$\mu_{MEM} = N_{Cell}\mu_{Cell} \text{ and } \sigma_{MEM} = \sqrt{N_{Cell}}\,\sigma_{Cell} \quad (6)$$

Fig. 3a shows the distribution of the leakage of a single SRAM cell shifted to different inter-die corners, due to random intra-die variation. Fig. 3b and 3c shows that the distribution of the memory leakage at different temperature, estimated using (6), which closely follows the Monte-Carlo simulation results. Compared with Fig. 3a, it is observed from Fig. 3b and Fig. 3c that by monitoring the whole memory leakage, instead of single cell leakage, the inter-die process corner of the memory chip can be identified with considerable margin.

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TABLE-I: Miss-prediction Error (%) (with Δ=5mV)						
Miss-prediction Type	Vt Boundary	STD. of Intra-Die Variation				
		40mV	50mV	60mV	70mV	80mV
P_{MISLVT}	25mV	0	0	0.006%	0.62%	7.%
	50mV	0	0	0.014%	0.61%	7.5%
	100mV	0	0	0.002%	0.72%	7.98%
P_{MISHVT}	25mV	0	0	0.008%	0.52%	6.46%
	50mV	0	0	0.008%	0.51%	6.02%
	100mV	0	0	0	0.42%	5.7%

E. Self-Repairing SRAM using Leakage Monitoring

In a Self-repairing SRAM using "Leakage Monitoring", the leakage (memory leakage) of the SRAM die is monitored using an online leakage monitor. The measured leakage is then compared with the reference currents to identify the inter-die process corner of the chip. Based on this measurement, the right body bias is applied to the chip. The schematic of a self-repairing SRAM array with self-adjustable body-bias generator is shown in Fig. 4. A simple current mirror circuit is used for leakage monitoring (Fig. 5). The transistors in the current mirror are designed to be large to reduce the effect of variations in the monitor circuit. The monitor generates an output voltage (Vout) that is proportional to the leakage of the SRAM array (Fig. 5). The output of the leakage monitor is compared with the reference voltages corresponding to the different inter-die process corners. The reference voltages can be generated using band-gap voltage sources [13]. Based on the results of this comparison, the body bias generator applies the right body bias to the SRAM array. If an SRAM die is in the low inter-die Vt corner, the output of the leakage monitor (Vout) will be greater than both the reference voltages (V_{REF1} and V_{REF2}) and both comparators generate zero, resulting in application of a reverse body bias (RBB). If the SRAM chip is in the high Vt corner, Vout of the leakage monitor will be less than both V_{REF1} and V_{REF2}. Hence, the outputs of the comparator will be at logic one. This results in application of a forward body bias (FBB). For SRAM dies that are in the nominal Vt corner, the leakage monitor output will be between V_{REF1} and V_{REF2} ($V_{REF1}>V_{REF2}$). Hence, the output of one the comparators will be at zero and the other one at one. This results in application of a zero body bias (ZBB=0). To avoid any performance loss due to the voltage drop across the leakage monitor (which is in the supply path) the leakage monitor is bypassed in the regular mode of operation. This bypassing is implemented by the PMOS bypass switch. Since the leakage monitor has to be bypassed when the memory is accessed, the outputs of the comparators are sampled to flip-flops (FF) by the "calibrate" signal. The repairing circuit starts operating when the calibrate signal is turned "on".

The self-repairing circuit shown in Fig. 4 is implemented in IBM 130nm CMOS (dual-Vt & triple-well) technology. Two 64KB SRAMs are implemented to emulate inter-die variations: one using low-Vt (representing extreme negative inter-die Vt shift) and the other using high-Vt (representing extreme positive inter-die Vt shift) devices. The on-chip leakage sensor is designed in series with the cell supply (not with pre-charge circuit) to monitor the array leakage. Large and distributed PMOS diodes are used in the current mirror to reduce the effect of within-die variability (~0.01% area overhead). The sensor output increases with array leakage and is compared against on-chip (or off-chip) reference voltages (representing low, nominal and high inter-die Vt corners) to apply proper body-bias (generated on/off chip). Level converters in the body-bias selection circuit convert logic "0" to the RBB value to prevent conduction through the 'off' NMOS transistors of pass gates connected to RBB voltage. The pass-gate devices are designed with thicker T_{ox} to prevent oxide breakdown when Vg='1' and Vs=RBB. The leakage monitors are bypassed (by switching "Calibrate" high-to-low) during regular operation using large PMOS devices (~5% area overhead). Access time is not impacted by the bypass PMOS as it is determined by the discharging current through access/pull-down NMOS.

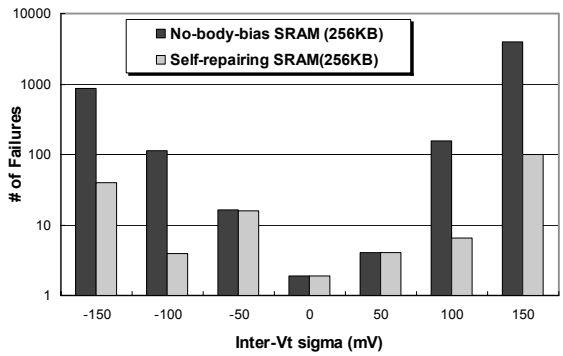


Fig. 6: Reduction in number of failures in 256KB memory array using self-repairing SRAM.

Let us now investigate the different sources of error in the application of the proper body bias in the SRAM array. To simplify the analysis let us only consider the errors that can occur for dies shifted to low inter-die Vt corners.

- First, due to intra-die variation the Vout for a low-Vt die with $\Delta Vt\text{-}inter = (-Vt_{bnd}-\Delta)$, (where, Δ is a small voltage ~5-10mV) may become lower than V_{REF1} which results in the application of ZBB (instead of RBB). This results in a miss-prediction in the low-Vt corner (P_{MISLVT}) and such a miss-prediction reduces the yield improvement using ABB. Similar, miss-prediction can occur for high-Vt dies with $\Delta Vt\text{-}inter = (Vt_{bnd} + \Delta)$. The probability of miss-prediction can be formally defined as:

$$P_{MISLVT} = P\left\{V_{out}\left(-Vt_{bnd}-\Delta\right) < V_{REF1}\right\}$$
$$P_{MISHVT} = P\left\{V_{out}\left(Vt_{bnd}+\Delta\right) > V_{REF2}\right\} \quad (10)$$

The miss-prediction error reduces with reduction in the intra-die Vt variation or increase in the memory size, which reduce the Vout spread (Table-I). Our simulation with 70nm BPTM shows that the error is less than 7% for 1KB memories and large within-die variation (Table-I). The error becomes negligible for memories larger than 2KB. However, it should be noted that this type of mis-prediction does not increase the number of faulty dies compared to the no-body-bias case. The mis-prediction error can be reduced by designing the Vt boundaries little inside the ZBB window (instead of exactly at the region boundaries).

- Let us consider a die in ZBB window (i.e. non-faulty one) and ΔVt-inter < 0. Due to presence of the intra-die variation in

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Vout and the "offset" voltage of the comparator (V_{off}), this die may be either detected as a HVT die or LVT die. If it is detected as a HVT die, FBB will be applied which will result in further reduction of its Vt, making the die a faulty one. We will refer to this event as *NegVt-FBB* and the probability of this error ($Y_{NegVt-FBB}$) is defined as:

$$P_{NegVt-FBB}(-\Delta Vt) = P\left(Vout(-\Delta Vt) < V_{REF2} + V_{off}\right) \times P_{MEM}\left(-\Delta Vt, V_b = FBB\right)$$

$P_{MEM}\left(-\Delta Vt, V_b = FBB\right)$ = Memory Failure Probability for a die with inter-die Vt= $-\Delta Vt$ and FBB

$$Y_{NegVt-FBB} = \sum_{\Delta Vt=0}^{Vt_{bnd}} P_{NegVt-FBB}(-\Delta Vt) \Big/ \#\text{ of dies within 0 and } -Vt_{bnd}$$

On the other hand, if it is detected as a LVT die, a RBB will be applied. Application of RBB to a die in the ZBB region may shift it to the HVT corner resulting in a faulty die. We will refer to this event as *NegVt-RBB* and the probability of this error ($Y_{NEGVT-RBB}$) is defined as:

$$P_{NegVt-RBB}(-\Delta Vt) = P\left(Vout(-\Delta Vt) > V_{REF1} - V_{off}\right) \times P_{MEM}\left(-\Delta Vt, V_b = RBB\right)$$

$$Y_{NegVt-RBB} = \sum_{\Delta Vt=0}^{Vt_{bnd}} P_{NegVt-RBB}(-\Delta Vt) \Big/ \#\text{ of dies within 0 and } -Vt_{bnd}$$

Similar arguments hold for the dies in the ZBB region with ΔVt-inter > 0. For those dies, the Vout variation and comparator offset may result in the application of either RBB (resulting in a higher Vt, referred to as *PosVt-RBB*) or FBB (resulting in a lower Vt, referred to as *PosVt-FBB*). The probability of occurrence of *NegVt-FBB* and *NegVt-RBB* increases with an increase in the intra-die variation and comparator offset voltage, and reduction in the separation between the Vt boundaries. However, our simulation result shows that, even with a comparator offset of 40mV and standard deviation of intra-die variation of 100mV this error is negligible (<0.01%). The low error due to *NegVt-FB* is due to the fact that $P\left(Vout(-\Delta Vt) < V_{REF2} + V_{off}\right)$ is negligible. This probability increases as ΔVt approached 0. However, as ΔVt$\rightarrow$0, the $P_{MEM}\left(-\Delta Vt, V_b = FBB\right)$ reduces since the die shifts further away from low-Vt corner. Hence, the product of the above two probabilities remain negligible in the entire negative ΔVt region of the ZBB window. Due to similar reason, errors due to *NegVt-RBB, PosVt-RBB* and *PosVt-FB* are also negligible.

Fig. 6 shows the reduction of number of failures in 256KB memory array shifted to different inter-die corners using self-repairing technique described above for 70nm predictive technology.

IV. Conclusions

In this paper, we discuss the failure mechanisms of SRAM due to process variations. The models for SRAM failure probabilities are introduced. SRAM test is optimized to cover all the failures resulted from process variations. In addition, two process tolerant design techniques are introduced. The proposed self-repairing SRAM circuit technique applies adaptive body bias to the SRAM array so as to minimize its failure probability. In the proposed technique, SRAM array leakage is monitored so as to determine different SRAM dies in appropriate inter-die Vt corners. Adaptive body bias is applied to the dies in the different Vt corners, resulting in a significant reduction in failures. The online monitoring of leakage is observed to be efficient in identifying the inter-die Vt corners of the SRAM dies. We believe with process variations becoming more and more pronounced in the scaled technology generation, the proposed fault tolerant design techniques will be useful to improve the yield of SRAM design.

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